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Asymmetric Catalysis

# Organocatalytic Enantioselective 1,12-Addition of Alkynyl Biphenyl Quinone Methides Formed In Situ

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**Abstract:** The chemistry of quinone methides formed in situ has been flourishing in recent years. In sharp contrast, the development and utilization of biphenyl quinone methides are rare. In this study, we achieved a remote stereocontrolled 1,12-conjugate addition of biphenyl quinone methides formed in situ for the first time. In the presence of a suitable chiral phosphoric acid, alkynyl biphenyl quinone methides were generated from α-[4-(4-hydroxyphenyl)phenyl]propargyl alcohols, followed by enantioselective 1,12-conjugate addition with indole-2-carboxylates. The strategy enabled the alcohols to serve as efficient allenylation reagents, providing practical access to a broad range of axially chiral allenes bearing a (1,1′-biphenyl)-4-ol unit, which were previously less accessible. Combined with control experiments, density functional theory calculations shed light on the reaction mechanism, indicating that enantioselectivity originates from the nucleophilic addition of alkynyl biphenyl quinone methides. Notably, not only the presence of biphenyl quinone methides as versatile intermediates was confirmed but also organocatalytic enantioselective 1,12-addition was established.

Over the past decade, quinone methides (QMs) have been identified as practical and high-value-added intermediates for the synthesis of functionalized molecules.[1] Notably, with the aid of chiral phosphoric acids (CPAs),[2] stereoselective reactions including nucleophilic addition, annulation and one-pot cascade reaction of QMs generated in situ from the corresponding hydroxybenzyl alcohols have been well-established for diversified and directed asymmetric synthesis (Scheme 1A).[3] In particular, the conjugate addition of these QMs, catalyzed by CPAs, offers a robust method for accessing axially chiral tetrasubstituted allenes.[4] However, organocatalytic stereoselective reactions of their extended analogues with carbonyl and methide units incorporated in different aromatic rings remain rare. Recently, Sun et al. have elegantly disclosed a CPA-catalyzed 1,6-conjugate addition of 2-naphthoquinone-8-methides (2-Nap-Q-8-Ms) formed in situ from 8-(hydroxyarylmethyl)naphthalen-2-ols (Scheme 1B).[5] At the same time, we independently developed a CPA-catalyzed 1,8-conjugate addition of 2-naphthoquinone-6-methides (2-Nap-Q-6-Ms) formed in situ from 6-(hydroxydiarylmethyl)naphthalen-2-ols for the synthesis of chiral tetraarylmethanes.[6] More importantly, the utilization of these versatile naphthoquinone methides (Nap-QMs) has provided a novel platform for the construction of enantioenriched complex target molecules.

As another type of extended QMs, biaryl quinone methides (BQMs) have been photogenerated from the corresponding 4-(4′-hydroxyphenyl)benzyl alcohols,[7] which cannot be readily generated via thermal chemistry. In particular, the application of BQMs as intermediates in the asymmetric synthesis has remained unknown (Scheme 1C). Obviously, the catalytic asymmetric transformation of such BQMs presents several challenges. Firstly, it is crucial to efficiently generate these BQMs in situ from the corresponding alcohols under mild conditions. Secondly, achieving selectivity in terms of chemo- and regioselectivity is a pivotal concern in the reactions of BQMs.[8] Thirdly, a formidable challenge arises in terms of stereocontrol for the remote addition of more extensively conjugated systems. In fact, organocatalytic remote stereocontrolled conjugate addition are still in its infancy.[9] Particularly, only three reports on organocatalyzed enantioselective 1,10-addition have been disclosed.[10] To our best knowledge, the organocatalytic enantioselective 1,12-addition has not been developed so far. To fill this gap, the development of an organocatalytic stereoselective 1,12-addition of BQMs formed in situ from α-[4-(4-hydroxyphenyl)phenyl]propargyl alcohols is urgent and in great demand.

Based on our previous work on organocatalytic stereoselective reactions of propargyl alcohols,[8d,10b,11] we first calculated the energy barrier for the dehydration assisted by dimethyl phosphate using density functional theory (DFT). As shown in Scheme 1D, the energy barrier for dehydration of 4′-(1-hydroxy-1,3-diphenylprop-2-yn-1-yl)-[1,1′-biphenyl]-4-ol to form alkynyl 4,4′-BQMs is 30.1 kcal/mol, indicating that the in situ formation of alkynyl 4,4′-BQM under mild

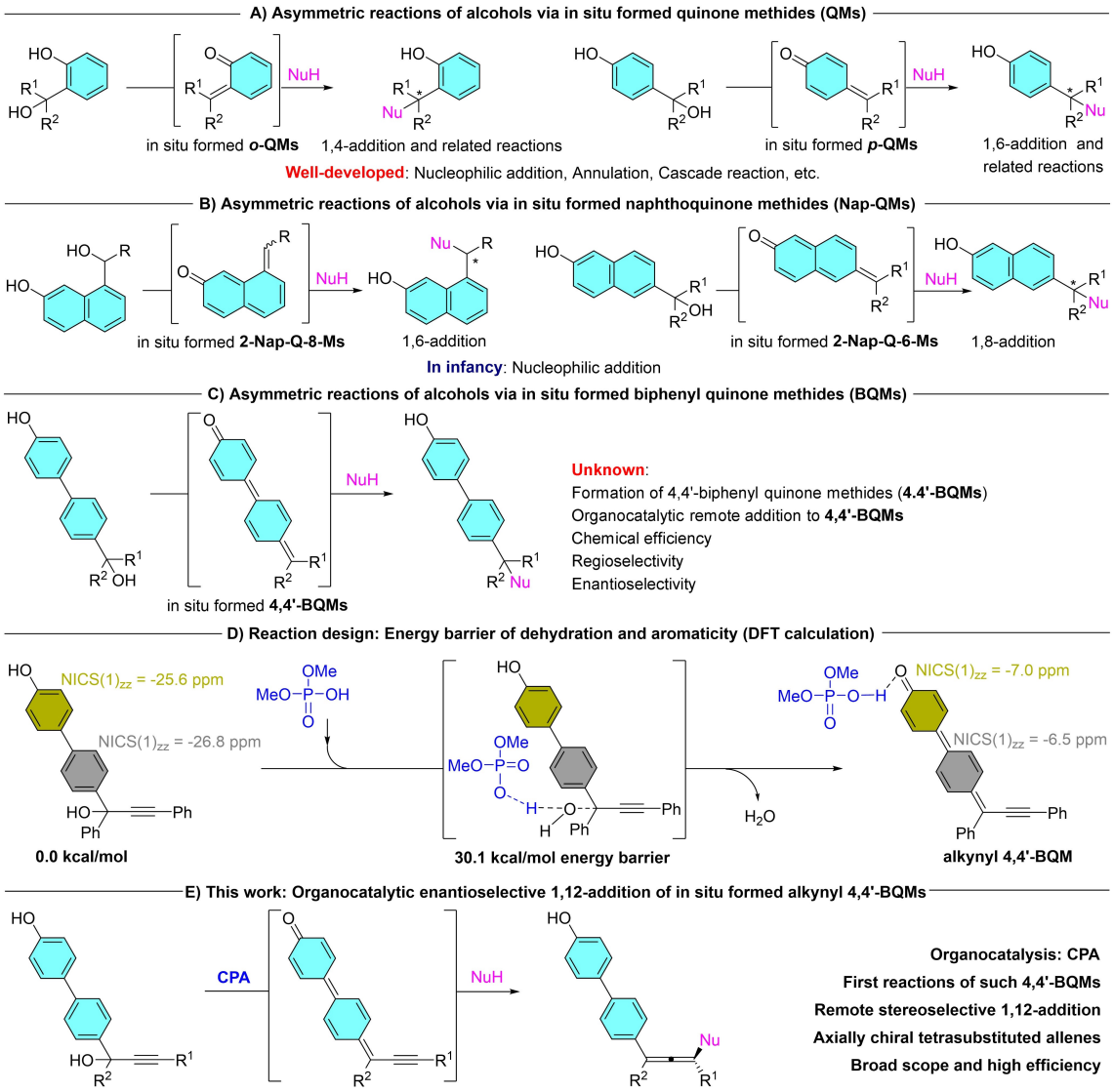
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***Scheme 1.*** Reactions of alcohols.

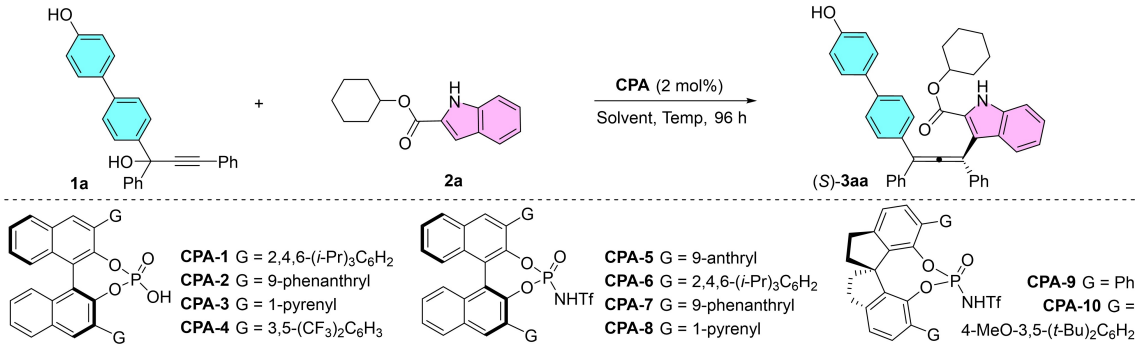
conditions is feasible. Meanwhile, based on nucleus-independent chemical shift (NICS) calculations,[12] the calculated NICS(1)$_{zz}$ values of biphenyl rings in propargyl alcohol are −25.6 and −26.8 ppm, respectively. In contrast, the calculated NICS(1)$_{zz}$ values of biphenyl rings in alkynyl 4,4′-BQM are −7.0 and −6.5 ppm. These results show that an apparent decrease in aromaticity of biphenyl rings from propargyl alcohol to alkynyl 4,4′-BQM, and rearomatization of alkynyl 4,4′-BQM would provide strong driving force for the subsequent reaction. In addition, the use of an appropriate catalytic system, reaction partners, and optimal conditions is crucial for achieving successful stereocontrol. Accordingly, we herein reported a chiral phosphoric acid catalyzed reaction of α-[4-(4-hydroxyphenyl)phenyl]propargyl alcohols via asymmetric 1,12-addition of in situ formed alkynyl 4,4′-BQMs (Scheme 1E). Notably, this work validated the ability of alkynyl 4,4′-BQMs to act as versatile intermediates in organocatalytic asymmetric reactions featuring remote stereocontrolled 1,12-conjugate addition for the synthesis of axially chiral tetrasubstituted allenes.[4a,13]

At the beginning of the investigation, we focused on CPAs as a result of their demonstrated excellent performance in the conjugate addition of electron-deficient systems formed in situ from the corresponding functionalized alcohols.[3–6,8,10–11,14–17] As shown in Table 1, the **CPA-1** catalyzed reaction between the model substrates 1-[4-(4-hydroxyphenyl)phenyl]-1,3-diphenylprop-2-yn-1-ol **1a** and cyclohexyl 1*H*-indole-2-carboxylate **2a** furnished allene **3aa** in 54 % yield (Table 1, entry 1). Although racemic product was obtained, it confirmed the effectivity of phosphoric acid. To improve the enantioselectivity, a series of chiral phosphoric acids were investigated (Table 1, entries 2–5). Specifically, the desired product (*S*)-**3aa** was obtained in 70 % yield with 40 % ee in the presence of catalyst **CPA-3** (Table 1, entry 3). Furthermore, catalyst **CPA-5** enabled the formation of the target product (*S*)-**3aa** in 83 % yield with 42 % ee (Table 1, entry 5). To our delight, enantioselectivity


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*Table 1:* Conditions optimization.




| Entry[a] | Catalyst | Solvent | Isolated yield [%] | ee [%][b] |
|---|---|---|---|---|
| 1 | **CPA-1** | $CH_2Cl_2$ | 54 | 0 |
| 2 | **CPA-2** | $CH_2Cl_2$ | 69 | 24 |
| 3 | **CPA-3** | $CH_2Cl_2$ | 70 | 40 |
| 4 | **CPA-4** | $CH_2Cl_2$ | 61 | 8 |
| 5 | **CPA-5** | $CH_2Cl_2$ | 83 | 42 |
| 6 | **CPA-5** | $CCl_4$ | 87 | 73 |
| 7 | **CPA-5** | $CCl_4$ (2.0 mL) | 82 | 82 |
| 8 | **CPA-6** | $CH_2Cl_2$ | 88 | 0 |
| 9 | **CPA-7** | $CH_2Cl_2$ | 82 | 52 |
| 10 | **CPA-8** | $CH_2Cl_2$ | 71 | 63 |
| 11 | **CPA-8** | toluene (1.0 mL) | 88 | 84 |
| 12 | **CPA-9** | $CH_2Cl_2$ | 51 | 0 |
| 13 | **CPA-10** | $CH_2Cl_2$ | 58 | 0 |

[a] Unless noted, a mixture of **1a** (0.05 mmol), **2a** (0.06 mmol) and **CPA** (2 mol%) in the solvent (0.3 mL) was stirred at 25 °C for 96 h. [b] Determined by HPLC analysis on a chiral stationary phase.

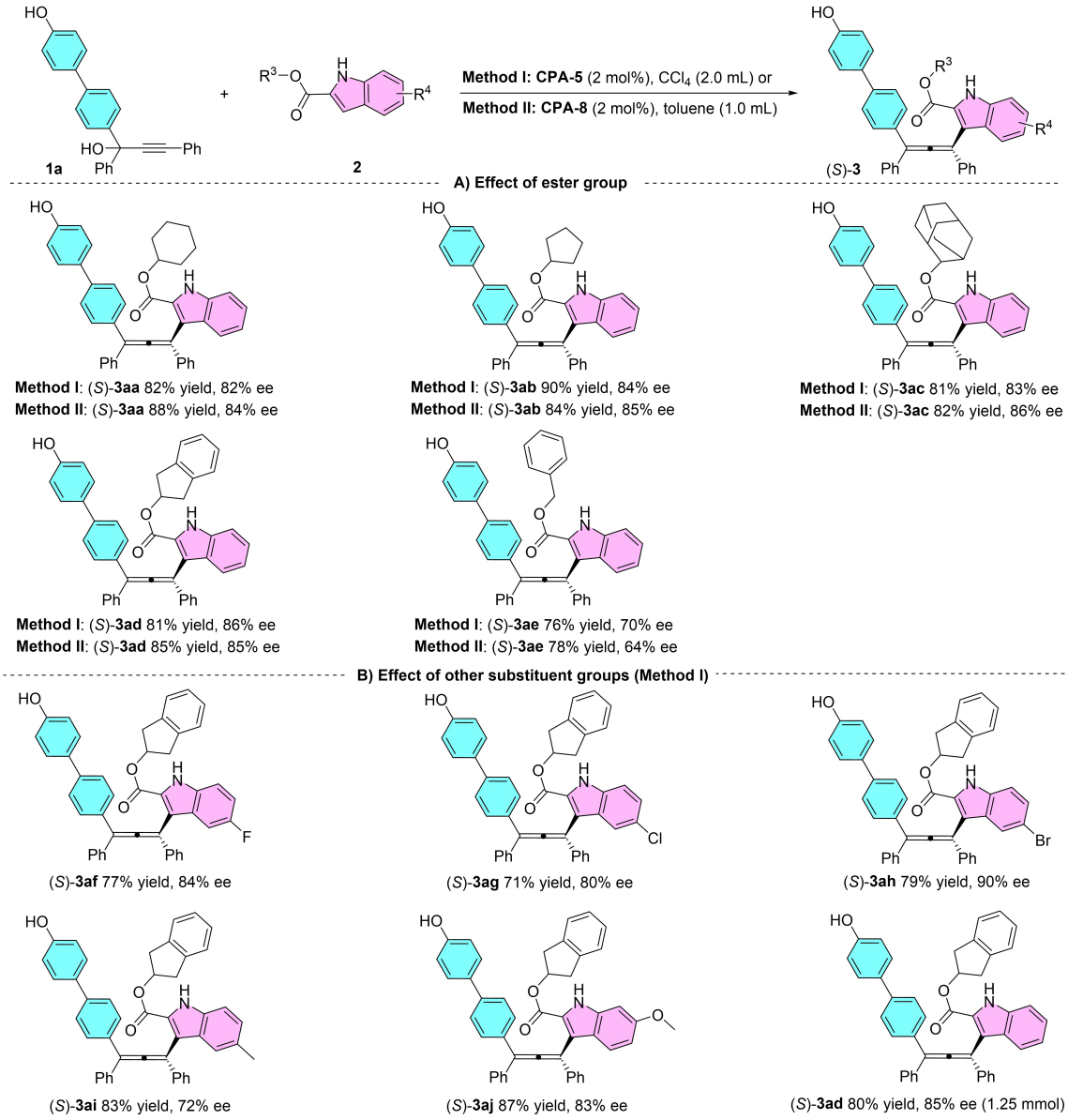
enhancement was achieved after the examination of reaction media, affording axially chiral allene (*S*)-**3aa** in 87% yield with 73% ee (Table 1, entry 6, see Supporting Information for more details). Further modification of reaction conditions enabled the formation of (*S*)-**3aa** in 82% yield with 82% ee (Table 1, entry 7, see Supporting Information for more details). To further improve the efficiency and stereoselectivity, we re-examined the catalytic system (Table 1, entries 8–10), disclosing that the desired product (*S*)-**3aa** was obtained in 71% yield with 63% yield from the **CPA-8** catalyzed reaction (Table 1, entry 10). Systematic screening studies including the effect of solvents, concentration, reaction temperature and various added additives revealed that **CPA-8** enabled the formation of (*S*)-**3aa** in 88% yield with 84% ee at 25 °C in toluene after 96 h (Table 1, entry 11, see Supporting Information for more details). When spiro catalyst **CPA-9** was employed, the racemic product was obtained in 51% yield (Table 1, entry 12). Similar results were obtained with spiro catalyst **CPA-10** (Table 1, entry 13).

With the established conditions, we surveyed the substrate scope of the reactions between propargyl alcohols and indole-2-carboxylates. In the presence of **CPA-5**, the effect of the ester group of indole-2-carboxylates was first investigated (Scheme 2A). Concretely, the ester group ($R^3$) was found to have a slight effect on the reaction, affording the corresponding products (*S*)-**3aa**–**ae** in 76–90% yields with 70–86% ee. Alternatively, the desired (*S*)-**3aa**–**ae** could be obtained in 78–88% yields with 64–86% ee with the aid of catalyst **CPA-8**. Furthermore, various substituents ($R^4$=F, Cl, Br, Me, MeO) could be introduced to the indole ring of indole-2-carboxylates and the desired (*S*)-**3af**–**aj** were obtained in 71–87% yields with 72–90% ee. Neither significant electronic nor steric effect of the aromatic moiety was observed. In all, a wide range of indole-2-carboxylates reacted smoothly to generate the corresponding axially chiral allenes **3** in high yields and enantioselectivities. To explore the synthetic potential, the **CPA-5** catalyzed reaction was scaled up to 1.25 mmol that furnished the corresponding product (*S*)-**3ad** with equally high efficiency and stereoselectivity as the small-scale reaction (80% yield, 85% ee).

With these encouraging data in hand, we turned our attention to the scope of α-[4-(4-hydroxyphenyl)phenyl]propargyl alcohols. As shown in Scheme 3, a broad scope of alcohols **1** bearing $R^1$/$R^2$ substituents was compatible to afford the desired axially chiral allenes in generally good to high yields and stereoselectivities. More specifically, a series of propargyl alcohols with aromatic substituent ($R^1$) were successfully applied in the **CPA-5** catalyzed reactions of indole-2-carboxylate **2d**, affording the target compounds (*S*)-**3bd**–**id** in 73–88% yields with 75–86% ee, respectively. Both electronic nature and steric hindrance of substituents have a little effect on the reaction. The aliphatic substituted propargyl alcohols ($R^1$ = *n*-Bu, cyclopropyl) were found to react smoothly to furnish the corresponding product (*S*)-**3jd** in 61% yield with 80% ee and (*S*)-**3kd** in 63% yield with 82% ee, respec-


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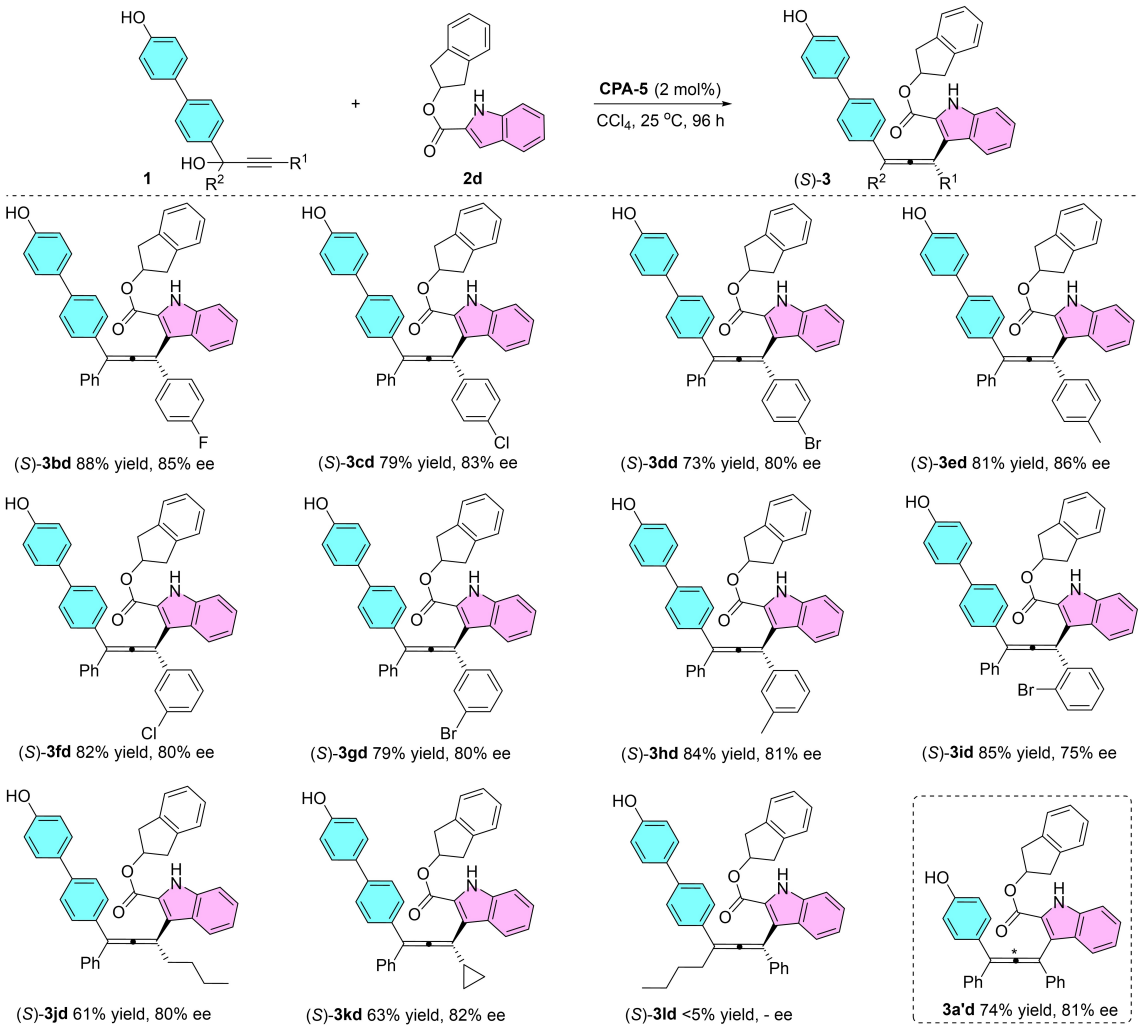

***Scheme 2.*** Scope of the reaction with respect to the 1*H*-indole-2-carboxylate **2**. A mixture of **1a** (0.05 mmol), **2** (0.06 mmol) and **CPA** (2 mol%) in the solvent was stirred at 25 °C for 96 h. Yields are for the isolated product **3**. The ee value was determined by HPLC analysis on a chiral stationary phase.

tively. As an exception, propargyl alcohol **1l** ($R^1$=Ph, $R^2$=*n*-Bu) was not tolerated under the standard conditions. Notably, under standard conditions, the asymmetric 1,8-conjugate addition of indole-2-carboxylate **2d** to alkynyl quinone methides formed in situ from 4-(1-hydroxy-1,3-diphenylprop-2-yn-1-yl)phenol **1a′** furnished the desired allene **3a′d** in 74% yield with 81% ee, which indicated that the chiral phosphoric acid-catalyzed conjugate addition was successfully expanded from 1,8-addition to 1,12-addition with the aid of (1,1′-biphenyl)-4-ol unit. Taken altogether, these results not only verified that the (1,1′-biphenyl)-4-ol unit served as a practical functional group to enable the formation of axially chiral allenes from propargyl alcohols, but also indicated that organocatalytic remote stereocontrol was achieved via the **CPA-5** catalyzed 1,12-conjugate addition of alkynyl 4,4′-BQMs formed in situ.

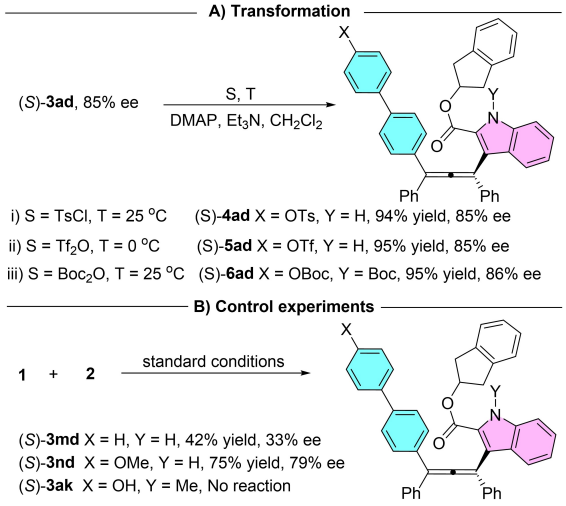
Esterification of free hydroxyl groups of axially chiral allenes (*S*)-**3ad** was investigated with several classical reagents (Scheme 4A). Notably, the corresponding products (*S*)-**4ad**, (*S*)-**5ad** and (*S*)-**6ad** were obtained in high efficiency without compromising the enantioselectivity. To further examine the effect of (1,1′-biphenyl)-4-ol unit, control experiments were surveyed (Scheme 4B). When α-[4-(4-phenyl)phenyl]propargyl alcohol **1m** was employed to react with indole-2-carboxylate **2d** under the standard conditions, allene (*S*)-**3md** was obtained in 42% yield with 33% ee. The reason might be that the absence of alkynyl biphenyl quinone methide formation likely contributed to the reaction proceeding via the $S_N1$ mechanism. When the


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***Scheme 3.*** Scope of the reaction with respect to the 1-(4-(4-hydroxylphenyl)phenyl) propargyl alcohol **1**. A mixture of **1** (0.05 mmol), **2d** (0.06 mmol) and **CPA-5** (2 mol%) in $CCl_4$ (2.0 mL) was stirred at 25 °C for 96 h. Yields are for the isolated product **3**. The ee value was determined by HPLC analysis on a chiral stationary phase.




***Scheme 4.*** Further investigations.

free hydroxyl group was blocked by methyl, the corresponding product (*S*)-**3nd** was obtained in 75% yield with 79% ee, which further confirmed the importance of oxygen atom of (1,1′-biphenyl)-4-ol unit. Moreover, no reaction occurred between propargyl alcohol **1a** and 1-methylindole-2-carboxylate **2l**, disclosing that the free amino group on indole-2-carboxylate were essential for controlling the efficiency and stereoselectivity.

The absolute configuration of product **3aa** was assigned by comparison between its experimental and calculated electronic circular dichroism (ECD) spectra to be the *S* configuration (Figure 1). Furthermore, kinetic studies indicated that this reaction is first order in propargylic alcohol **1a** (Figure 2A), first order in chiral catalyst **CPA-5** (Figure 2B), and order of about 0.7 in nucleophile **2f** (Figure 2C).

To better understand the mechanism and origins of stereoselectivity, DFT calculations were carried out upon **CPA-5** mediated reaction between alcohol **1a** and indole-2-carboxylate **2d** (see Supporting Information for details). As


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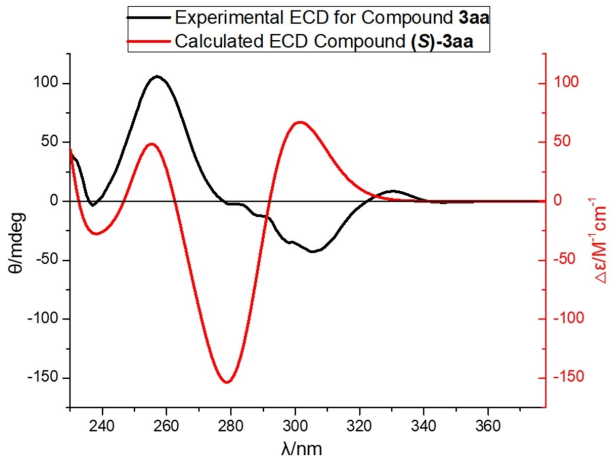



***Figure 1.*** Comparison of the calculated ECD spectrum of compound (*S*)-**3aa** with the experimental ECD spectrum of compound **3aa**.

shown in Figure 3, the reaction was initiated by the formation of hydrogen bond complex **int-1** from substrate **1a** and chiral phosphoric acid **CPA-5** with an endergonic process of 3.9 kcal/mol. The intermediate **int-1** then undergoes dehydration via transition state **TS1** to afford an identical intermediate alkynyl BQM **int-2** with an energy barrier of 18.8 kcal/mol. With the participation of indole-2-carboxylate nucleophile **2d**, the catalyst could protonate BQM and form a hydrogen-bonded complex **int-3*S***. Subsequently, the intermolecular nucleophilic addition via transition state **TS2-*S*** with a corresponding energy barrier of 13.8 kcal/mol generates the intermediate **int-4*S***. Finally, intermediate **int-4*S*** could undergo rearomatization/proton transfer via transition state **TS3-*S*** with an energy barrier of 5.3 kcal/mol, generating the product (*S*)-**3ad** and regenerating the catalyst **CPA-5**. The computational results disclosed the intermolecular nucleophilic addition of indole-2-carboxylate **2d** to alkynyl BQM **int-2** is considered to be the enantio- and rate-determining step for this reaction. It is noteworthy that this step requires a smaller activation energy than the preceding dehydration step in terms of energy of single steps. However, due to the high-energy intermediate **int-3**, the transition state **TS-2** for this step becomes the highest-energy point on the computed free-energy profile. The rate-determining step involves the reversibly generated high-energy BQM intermediate, the catalyst, and the indole-2-carboxylate, which is consistent with results from the kinetic studies. Regarding the stereochemistry, the intermolecular nucleophilic addition via transition state **TS2-*R*** with an energy barrier of 15.3 kcal/mol would generate intermediate **int-4*R***, which is the precursor of the product (*R*)-**3ad**. The relative free energy of the transition state **TS2-*R*** is 2.0 kcal/mol higher than that of transition state **TS2-*S***, which indicates that the generation of the *R*-configured product is unfavorable. The calculated enantiomeric excess (ee) is 93% based on the energy difference between transition states **TS2-*R*** and **TS2-*S***, which slightly overestimated the experimental result (86% ee). Furthermore, optimized structures of transition states **TS2-*S*** and **TS2-*R*** were compared (Figure 4A). Notably, the N–H···O–S and C–H···O–P hydrogen-bonding interactions existed in stereoisomeric transition states **TS2-*R*** and **TS2-*S***. Particularly, transition state **TS2-*S*** enjoys additional stabilizing C–H···O–P hydrogen-bonding interactions with contact distances of 2.21 and 2.30 Å, respectively. These more stable hydrogen-bonding interactions contributed significantly to the stabilization of transition state **TS2-*S***, which is likely the most important factor that determining the stereoselectivity. In contrast, there is a lack of additional stabilizing C–H···O–P hydrogen-bonding interactions in the disfavored transition state **TS2-*R***. Further calculations by removing the substituents of the catalyst with methyl groups, then computing the single-point energy difference $\Delta\Delta E^{\ddagger}$ without optimization. The calculated results show that transition state **TS2-*S*** is still 2.1 kcal/mol lower in energy than transition state **TS2-*R***, indicating that hydrogen-bonding interactions play a leading role in achieving high enantioselectivity. To further investigate the origin of stereoselectivity, we performed the quantitative Bader's atoms-in-molecules (AIM)[18] analyses for the transition states **TS2-*S*** and **TS2-*R***. The value of Laplacian of electron densities ($\nabla^2\rho$) could be used to quantitatively measure the strength of the weak interaction. As depicted in Figure 4B/C, four C–H···O (2.06, 2.10, 2.21 and 2.30 Å) and one N–H···F (1.73 Å) interactions can be found in **TS2-*S***, while two C–H···O (2.04 and 2.29 Å) and one N–H···F (1.64 Å) interactions exist in **TS2-*R***. Taking the above into consideration, the C–H···O hydrogen-bond interactions in **TS2-*S*** are stronger than those observed in **TS2-*R***. Therefore, the **TS2-*S*** is more stable than **TS2-*R***, and the corresponding pathway associated with **TS2-*S*** has a lower energy barrier. To further verify the accuracy of our computational results, single-point energies were also

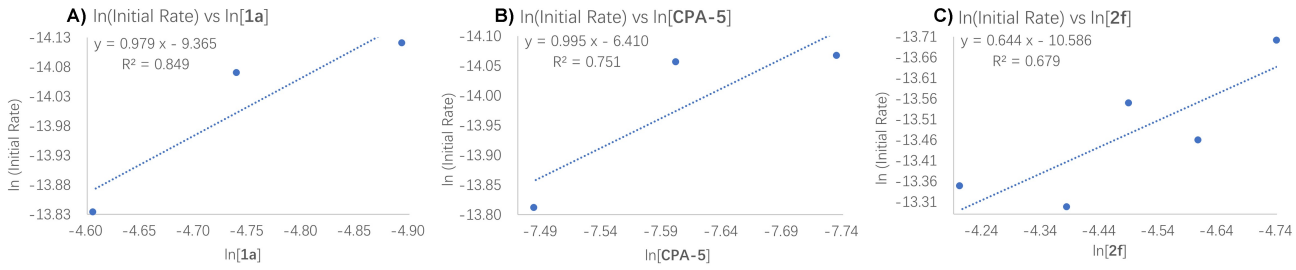



***Figure 2.*** Kinetic studies. A) Kinetic study of propargylic alcohol **1a**. B) Kinetic study of **CPA-5**. C) Kinetic study of indole-2-carboxylate **2f**.


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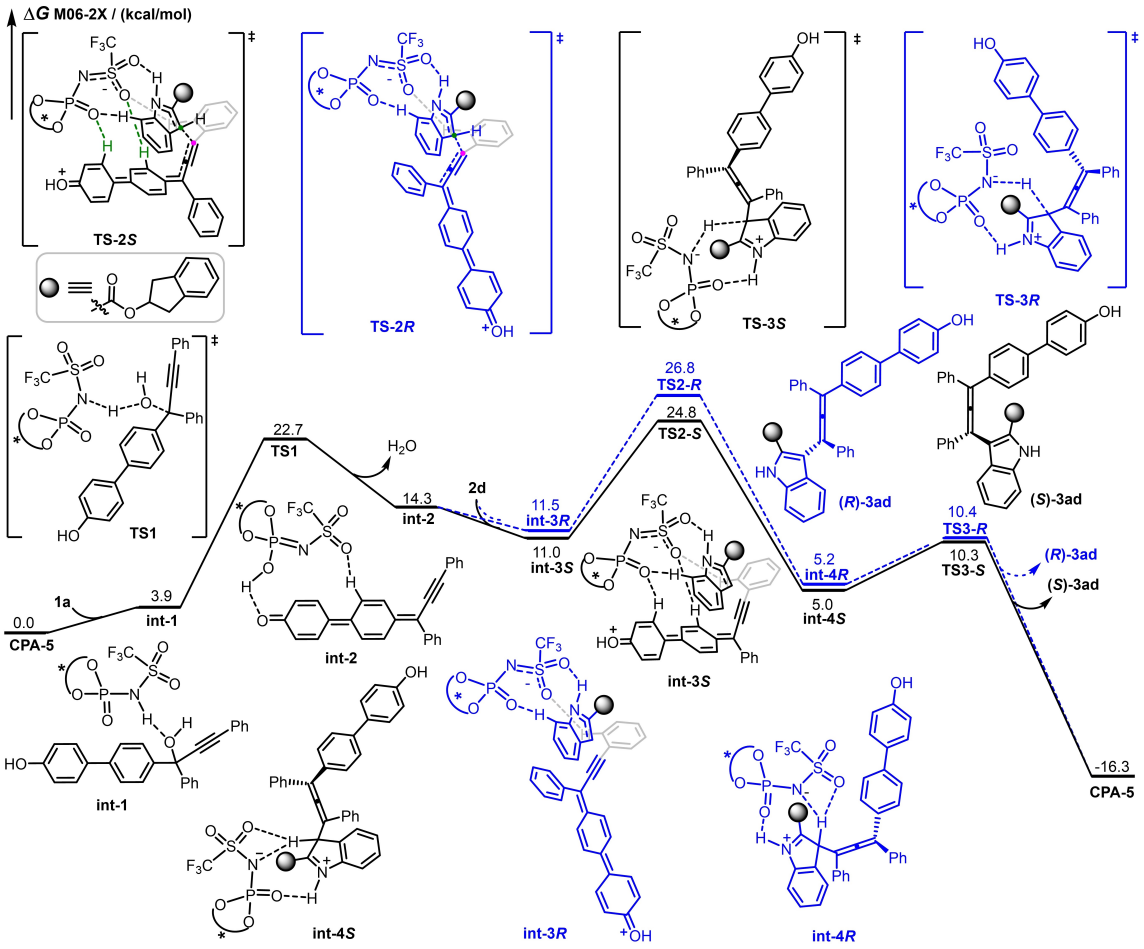



***Figure 3.*** DFT study. Free energy profile for the **CPA-5**-catalyzed reaction of **1a** with **2d**. The energy values are in given kcal/mol and represent the relative free energies calculated at the M06-2X/6-311+G(d,p)-SMD($CCl_4$)//B3LYP-D3(BJ)/6-31G(d) level of theory. The distances are given in Å.

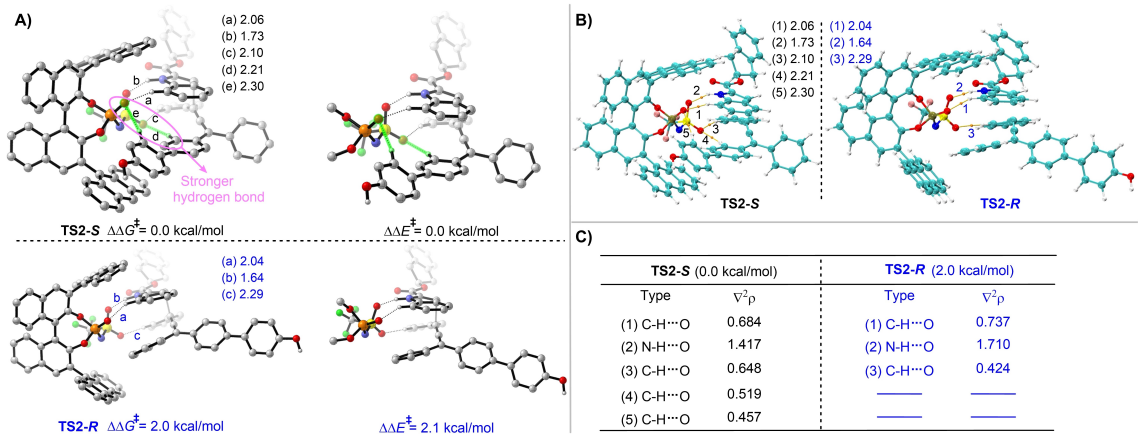



***Figure 4.*** DFT study. Origin of the enantioselectivity. A) The geometry information of transition states **TS2-*S*** and **TS2-*R***. B) AIM analyses of the transition states **TS2-*S*** and **TS2-*R***. C) Summary of Laplacian values ($\nabla^2\rho$ in eÅ$^{-3}$). The energy values are in given kcal/mol and represent the relative free energies calculated at the M06-2X/6-311+G(d,p)-SMD($CCl_4$)//B3LYP-D3(BJ)/6-31G(d) level of theory. The distances are given in Å.

evaluated within the SMD model using the ωB97X-D, MN15, M11 and PBE0-D3BJ functionals to compare the stereoselectivities computed using different methods (see Supporting Information for details, Table S3). The predicted enantioselectivities are consistent with each of the five density functional methods.

In conclusion, while quinone methides have been extensively studied, the field of stereoselective organo-


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catalytic reactions involving their extended analogues, biphenyl quinone methides, remains unexplored. Herein, we have established such a reaction for the first time. In detail, alkynyl biphenyl quinone methides were generated in situ from α-[4-(4-hydroxyphenyl)phenyl]propargyl alcohols with the aid of a chiral phosphoric acid catalyst, followed by enantioselective 1,12-conjugate addition of indole-2-carboxylates to furnish a wide range of axially chiral tetrasubstituted allenes with high efficiency. Control experiments and DFT calculations were utilized to investigate the reaction pathway. Through hydrogen-bonding interactions, a new organocatalytic remote stereocontrol model was developed. Of note, not only alkynyl biphenyl quinone methides formed in situ were developed as versatile intermediates for asymmetric synthesis, but also organocatalytic enantioselective 1,12-additions were achieved for the construction of axially chiral tetrasubstituted allenes. Additional studies on biphenyl quinone methides are underway in our laboratories.

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## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

**Keywords:** 1,12-addition · axial chiral allenes · chiral phosphoric acids · biphenyl quinone methides · propargyl alcohols

[1] For selected reviews, see: a) R. W. Van De Water, T. R. R. Pettus, *Tetrahedron* **2002**, *58*, 5367–5405; b) N. J. Willis, C. D. Bray, *Chem. Eur. J.* **2012**, *18*, 9160–9173; c) S. Gnaim, D. Shabat, *Acc. Chem. Res.* **2014**, *47*, 2970–2984; d) Q. Li, T. Dong, X. Liu, X. Zhang, X. Yang, X. Lei, *Curr. Org. Chem.* **2014**, *18*, 86–92.

[2] For pioneering examples of CPA catalysis, see: a) T. Akiyama, J. Itoh, K. Yokota, K. Fuchibe, *Angew. Chem. Int. Ed.* **2004**, *43*, 1566–1568; b) D. Uraguchi, M. Terada, *J. Am. Chem. Soc.* **2004**, *126*, 5356–5357; c) D. Nakashima, H. Yamamoto, *J. Am. Chem. Soc.* **2006**, *128*, 9626–9627; For selected reviews on CPA catalysis, see: d) D. Parmar, E. Sugiono, S. Raja, S. M. Rueping, *Chem. Rev.* **2014**, *114*, 9047–9153; e) T. Akiyama, K. Mori, *Chem. Rev.* **2015**, *115*, 9277–9306; f) T. James, M. van Gemmeren, B. List, *Chem. Rev.* **2015**, *115*, 9388–9409; g) J. Kikuchi, M. Terada, *Chem. Eur. J.* **2021**, *27*, 10215–10225.

[3] For reviews on asymmetric transformations of QMs, see: a) T. P. Pathak, M. S. Sigman, *J. Org. Chem.* **2011**, *76*, 9210–9215; b) L. Caruana, M. Fochi, L. Bernardi, *Molecules* **2015**, *20*, 11733–11764; c) Z. Wang, J. Sun, *Synthesis* **2015**, *47*, 3629–3644; d) A. A. Jaworski, K. A. Scheidt, *J. Org. Chem.* **2016**, *81*, 10145–10153; e) W. Li, X. Xu, P. Zhang, P. Li, *Chem. Asian J.* **2018**, *13*, 2350–2359; f) C. G. S. Lima, F. P. Pauli, D. C. S. Costa, A. S. de Souza, L. S. M. Forezi, V. F. Ferreira, F. de Carvalho da Silva, *Eur. J. Org. Chem.* **2020**, *2020*, 2650–2692; g) X. Li, Z. Li, J. Sun, *Nat. Synth.* **2022**, *1*, 426–438; h) C. Dorsch, C. Schneider, *Synthesis* **2022**, *54*, 3125–3141; i) H. Wang, S. Yang, Y. Zhang, F. Shi, *Chin. J. Org. Chem.* **2023**, *43*, 974–999.

[4] For a review, see: a) X. Wang, X. Chen, W. Lin, P. Li, W. Li, *Adv. Synth. Catal.* **2022**, *364*, 1212–1222. For selected examples, see: b) W.-F. Zheng, W. Zhang, C. Huang, P. Wu, H. Qian, L. Wang, Y.-L. Guo, S. Ma, *Nat. Catal.* **2019**, *2*, 997–1005; c) J. Yang, Z. Wang, Z. He, G. Li, L. Hong, W. Sun, R. Wang, *Angew. Chem. Int. Ed.* **2020**, *59*, 642–647; d) X. Li, J. Sun, *Angew. Chem. Int. Ed.* **2020**, *59*, 17049–17054; e) C. Qian, T. Huang, J. Sun, P. Li, *Org. Lett.* **2022**, *24*, 6472–6476.

[5] S. Liu, K. L. Chan, Z. Lin, J. Sun, *J. Am. Chem. Soc.* **2023**, *145*, 12802–12811.

[6] M. Liu, B. Shen, C. Liu, P. Yu, P. Li, *J. Am. Chem. Soc.* **2023**, *145*, 14562–14569.

[7] a) Y. Shi, P. Wan, *J. Chem. Soc. Chem. Commun.* **1995**, 1217–1218; b) Y. Shi, P. Wan, *Can. J. Chem.* **2005**, *83*, 1306–1323; c) M. Xu, M. Lukeman, P. Wan, *Photochem. Photobiol.* **2006**, *82*, 50–56; d) N. Basaric, N. Cindro, D. Bobinac, K. Mlinaric-Majerski, L. Uzelac, M. Kralj, P. Wan, *Photochem. Photobiol. Sci.* **2011**, *10*, 1910–1925.

[8] For selected examples on regio- and stereoselective addition of QMs, see: a) M. Chen, J. Sun, *Angew. Chem. Int. Ed.* **2017**, *56*, 11966–11970; b) D. Qian, L. Wu, Z. Lin, J. Sun, *Nat. Commun.* **2017**, *8*, 567; c) M. Chen, D. Qian, J. Sun, *Org. Lett.* **2019**, *21*, 8127–8131; d) F. Li, X. Chen, S. Liang, Z. Shi, P. Li, W. Li, *Org. Chem. Front.* **2020**, *7*, 3446–3451.

[9] a) D. Uraguchi, K. Yoshioka, Y. Ueki, T. Ooi, *J. Am. Chem. Soc.* **2012**, *134*, 19370–19373; b) T. den Hartog, Y. Huang, M. Fañanás-Mastral, A. Meuwese, A. Rudolph, M. Pérez, A. J. Minnaard, B. L. Feringa, *ACS Catal.* **2015**, *5*, 560–574. For a related review, see: c) Z. Yue, F. Fang, Y. Xia, P. Li, W. Li, *Adv. Synth. Catal.* **2023**, *365*, 1926–1933.

[10] a) X. Li, M. Duan, P. Yu, K. N. Houk, J. Sun, *Nat. Commun.* **2021**, *12*, 4881; b) X. Lin, B. Shen, Z. Wang, Y. Cheng, X. Chen, P. Li, P. Yu, W. Li, *Org. Lett.* **2022**, *24*, 4914–4918; c) Z. Wang, Y. Cheng, Z. Yue, X. Chen, P. Li, W. Li, *Asian J. Org. Chem.* **2022**, e202200399.

[11] a) P. Zhang, Q. Huang, Y. Cheng, R. Li, P. Li, W. Li, *Org. Lett.* **2019**, *21*, 503–507; b) L. Zhang, Y. Han, A. Huang, P. Zhang, P. Li, W. Li, *Org. Lett.* **2019**, *21*, 7415–7419; c) Z. Wang, X. Lin, X. Chen, P. Li, W. Li, *Org. Chem. Front.* **2021**, *8*, 3469–3474; d) Y. Wu, Z. Yue, C. Qian, X. Chen, F. Li, P. Li, W. Li, *Asian J. Org. Chem.* **2022**, *11*, e202100724; e) C. Qian, M. Liu, J. Sun, P. Li, *Org. Chem. Front.* **2022**, *9*, 1234–1240; f) C. Qian, T. Huang, J. Sun, P. Li, *Org. Lett.* **2022**, *24*, 6472–6476; g) Y. Xia, M. Liu, C. Qian, P. Li, M. Dong, W. Li, *Org.*



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*Chem. Front.* **2023**, *10*, 30–34; h) Z. Yue, B. Shen, J. Cao, X. Chen, F. Fang, P. Li, P. Yu, W. Li, *Org. Chem. Front.* **2023**, *10*, 3662–3668.

[12] Z. Chen, C. S. Wannere, C. Corminboeuf, R. Puchta, P. von R Schleyer, *Chem. Rev.* **2005**, *105*, 3842–3888.

[13] Q. H. Li, X. Jiang, K. Wu, R. Q. Luo, M. Liang, Z. H. Zhang, Z. Y. Huang, *Curr. Org. Chem.* **2020**, *24*, 694–708.

[14] For pioneering examples of CPA-catalyzed reactions of α-(3-indolyl)propargylic alcohols, see: a) M. Rueping, B. J. Nachtsheim, S. A. Moreth, M. Bolte, *Angew. Chem. Int. Ed.* **2008**, *47*, 593–596; b) Q.-X. Guo, Y.-G. Peng, J.-W. Zhang, L. Song, Z. Feng, L.-Z. Gong, *Org. Lett.* **2009**, *11*, 4620–4623; c) T. Liang, Z. Zhang, J. C. Antilla, *Angew. Chem. Int. Ed.* **2010**, *49*, 9734–9736. For a review, see: d) G.-J. Mei, F. Shi, *Chem. Commun.* **2018**, *54*, 6607–6621.

[15] For reviews on CPA-catalyzed reactions of α-(2-indolyl)propargylic alcohols, see: a) Y.-C. Zhang, F. Jiang, F. Shi, *Acc. Chem. Res.* **2020**, *53*, 425–446; b) H.-H. Zhang, F. Shi, *Acc. Chem. Res.* **2022**, *55*, 2562–2580; c) H. Zhang, F. Shi, *Chin. J. Org. Chem.* **2022**, *42*, 3351–3372.

[16] For pioneering examples of CPA-catalyzed reactions of α-(7-indolyl)propargylic alcohols, see: a) H.-Q. Wang, M.-M. Xu, Y. Wan, Y.-J. Mao, G.-J. Mei, F. Shi, *Adv. Synth. Catal.* **2018**, *360*, 1850–1860; b) C. Yue, F. Na, X. Fang, Y. Cao, J. C. Antilla, *Angew. Chem. Int. Ed.* **2018**, *57*, 11004–11008; c) F. Li, Z. Yang, Y. Yang, Q. Huang, X. Chen, P. Li, M. Dong, W. Li, *Adv. Synth. Catal.* **2021**, *363*, 2557–2561.

[17] For examples of CPA-catalyzed reactions of α-(6-indolyl)propargylic alcohols, see: a) Q. Song, P. Zhang, S. Liang, X. Chen, P. Li, W. Li, *Org. Lett.* **2020**, *22*, 7859–7863; b) X. Wang, Q. Song, X. Chen, P. Li, Y. Qi, W. Li, *Chin. J. Org. Chem.* **2022**, *42*, 1722–1734.

[18] R. F. W. Bader, *Chem. Rev.* **1991**, *91*, 893–928.

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